

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. `<media src="video.mp4"></media>`

- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. `<div>`
- B. `<nav>`
- C. `<section>`
- D. `<header>`

Snippet 4: XML Syntax

- A. `<note><to>User</to><from>Admin</from>`
- B. `<note><to>User</to><from>Admin</from></note>`
- C. `<note><to>User</from></note>`
- D. `<note to="User"></note>`

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I P Sai Bhargav (192411049) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sai Bhargav
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Scenario: Web Development Standards Implementation

1. Select the correct HTML tag for video embedding and justify your answer.

Correct Answer: B. <video src="video.mp4" controls></video>

Justification:

- The <video> element is the standard HTML5 tag used to embed videos in a webpage.
- The controls attribute provides built-in play, pause, volume, and fullscreen controls.
- It is supported by all modern web browsers.

Why the other options are incorrect:

- **A. <media>** – Not a valid HTML element.
- **C. <movie>** – Not a valid HTML5 tag.
- **D. <vid>** – Not a valid HTML tag.

2. Identify the appropriate HTTP method for form submission and explain why.

Correct Answer: B. POST

Justification:

- POST sends form data in the **request body**, making it more secure than GET.
- It is suitable for registration forms because it handles sensitive information like usernames, passwords, and email addresses.
- It supports large amounts of data and does not expose the data in the URL.

Why the other options are incorrect:

- **A. GET** – Sends data in the URL; not suitable for sensitive information.
- **C. PUT** – Used to update existing resources.
- **D. DELETE** – Used to delete resources.

3. Choose the proper semantic element for navigation and justify its usage.

Correct Answer: B. <nav>

Justification:

- The <nav> element represents a section containing navigation links.
- It improves website structure, accessibility, and search engine optimization (SEO).
- Screen readers can easily identify navigation menus.

Why the other options are incorrect:

- **A. <div>** – Generic container with no semantic meaning.
- **C. <section>** – Groups related content but is not specifically for navigation.
- **D. <header>** – Represents introductory content and may contain navigation, but it is not the dedicated navigation element.

4. Identify the correct XML syntax and explain the error in other options.

Correct Answer: B.

```
<note>
```

```
  <to>User</to>
```

```
  <from>Admin</from>
```

```
</note>
```

Justification:

- Every opening tag has a matching closing tag.
- The XML document is properly nested and well-formed.

Errors in the other options:

- **A.**
- `<note><to>User</to><from>Admin</from>`
 - Missing the closing `</note>` tag.
- **C.**
- `<note><to>User</from></note>`
 - Tags are mismatched (`<to>` is closed with `</from>`).
- **D.**
- `<note to="User"></note`
 - Missing the closing angle bracket (`>`) in the end tag, making it invalid XML.

5. Choose the correct protocol for web communication and justify your choice.

Correct Answer: B. HTTP

Justification:

- **HTTP (HyperText Transfer Protocol)** is the standard protocol used for communication between web browsers and web servers.
- It enables the transfer of web pages, images, videos, and other web resources.
- For secure communication, **HTTPS (HTTP Secure)** is commonly used, which encrypts data using SSL/TLS.

Why the other options are incorrect:

- **A. FTP** – Used for file transfer between computers.
- **C. SMTP** – Used for sending emails.
- **D. SNMP** – Used for network monitoring and management.

Summary of Correct Answers

Question	Correct Option	Answer
1	B	<video src="video.mp4" controls></video>
2	B	POST
3	B	<nav>
4	B	<note><to>User</to><from>Admin</from></note>
5	B	HTTP (HTTPS for secure web communication)